

WYSIWYG: What You See Is Where Your Gaze

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Abstract

As Picasso said, *a painting lives only through the one who looks at it*. To materialize this thought, we propose to automatically produce artworks that visually transform paintings by amplifying and distorting the most observed areas by viewers. Our work is based on a study conducted at the Caen Museum of Fine Arts in France. During the study, 151 participants were equipped with eye-tracking glasses, and observed various paintings, first alone and then in pairs. Based on the fixation and gaze path stored data, we first generate saliency maps that reflect the visual attention given to each painting. These maps are then used to fine-tune the UNETRSal model, a neural network designed to predict saliency maps, in order to align its outputs with human visual patterns observed during the experiment. The saliency maps generated are subsequently used to create deformations of the original painting. This overall process gives rise to a new artwork born from the interaction between human gaze and AI-prediction.

CCS Concepts

• **Computing methodologies** → **Interest point and salient region detections**; **Neural networks**; • **Human-centered computing** → **Mixed / augmented reality**.

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MM '25, Dublin, Ireland

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ACM ISBN 979-8-4007-2035-2/2025/10

<https://doi.org/10.1145/3746027.3756140>

Keywords

Eye-tracking, Saliency prediction, Visual attention, Gaze-driven generation, Art-based transformation.

ACM Reference Format:

Raphaëlle Lemaire*, Azamat Kaibaldiyeu*, Eléonore Mariette, Débora Viglieri, Alexis Lechervy, Fabrice Maurel, Gaël Dias, Jérémie Pantin, Gaëtane Blaizot, Véronique Agin, Nicolas Poirrel, Eric Bui, Hervé Platel, Denis Vivien, and Youssef Chahir. 2025. WYSIWYG: What You See Is Where Your Gaze. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746027.3756140>

1 Introduction

Art is a lie that makes us realize truth, said Picasso, capturing the tension between perception and interpretation that lies at the heart of visual experience. This interplay becomes especially compelling when computational models attempt to understand, or emulate, how humans engage with visual art [7]. Our study explores this idea by projecting human perception into 11 paintings to produce new artworks through targeted deformations of the original images. The project originates from a scientific study conducted at the Caen Museum of Fine Arts in France, which aims to investigate the cognitive and emotional benefits of museum visits. To this end, in addition to completing neuropsychological questionnaires, 151 visitors were equipped with eye-tracking glasses to record their visual trajectories and gaze points [2, 3]. The paintings included various types such as portraits, surrealism, pointillism and genre scenes. Based on the eye-tracking stored data, we generated saliency maps that model the visual attention paid to the paintings of the study. These maps were then used to fine-tune a saliency map predictor, called UNETRSal [6], enabling to anticipate which areas of a painting are most likely to attract human gaze. The new artwork emerges from visually transforming the original painting based on the saliency map generated by UNETRSal. The most observed zones

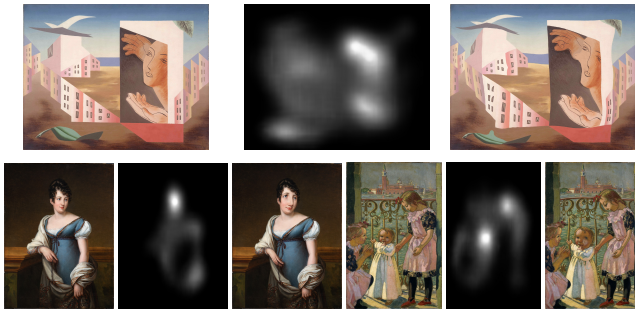


Figure 1: Three of the eleven paintings: the original, the predicted saliency maps, and the visual transformations.

are amplified, revealing the visual attention of all study participants (cf. Table 1). The demonstration video can be viewed [here](#).

2 Data Acquisition Protocol

We collected data from 151 participants equipped with eye-tracking glasses¹, during two visits to the Caen Museum of Fine Arts from September 2024 to May 2025. For the first visit, each participant viewed six paintings for 2 minutes. Two groups of visitors were randomly formed: a first group (A) with 74 people and another one (B) with 77 visitors, each group looking at three paintings in common and three other paintings exclusive to their group. Only group B received an explanation of the painting during the 2 minute observation phase. During the second visit, participants were paired, each pair composed of visitors from group A and group B. These pairs jointly observed eight paintings, including four in common with the first visit. A calibration of the eye-tracking glasses was performed before each visit. This step consisted of following the appearance of fixed points projected on a screen, without moving the head. This allowed the recording of gaze paths and fixation points for each participant during 2 minutes in front of the painting.

3 Saliency Map Prediction

We use the UNETRSal [6] model to predict generic saliency maps from the paintings viewed in the museum. UNETRSal is a hybrid model combining a Transformer-based encoder derived from UNETR [4] and a hierarchical convolutional decoder, which is specifically adapted to 2D image saliency prediction. While UNETR was initially designed for 3D medical image segmentation, its backbone was modified by replacing the 3D patch embeddings with 2D patch extraction, and by introducing a reshaping mechanism to reconstruct 2D saliency maps from Transformer outputs. The decoder is composed of multi-scale deconvolution and convolution blocks, where batch normalization layers were removed from specific blocks. This removal helped to stabilize training and improved saliency prediction. The model was trained using a composite loss function that combines Kullback-Leibler Divergence, Pearson's Correlation Coefficient, and Similarity metrics. This combination ensures alignment in distribution, linear correlation and structural similarity between predicted and ground-truth saliency maps.

UNETRSal has demonstrated strong performance on eye-tracking saliency benchmark datasets like SALICON [5] and CAT2000 [1].

¹The Core model from Pupil Labs available at <https://pupil-labs.com/products/core>.

We used the pretrained model on SALICON to fine-tune it using the eye-tracking data collected during museum visits in order to better align its predictions with the human gaze museum-based behavior. We chose the SALICON pretrained model as this is the largest available eye-tracking dataset, making it a strong foundation for pretraining. Indeed, it is commonly used as a basis for fine-tuning on smaller and task-specific datasets [6]. In order to fine-tune the SALICON-based UNETRSal model, we built a dataset that associates each painting with all participants individual ground-truth saliency maps. We also add a collective saliency map to each painting, by averaging the participants individual saliency maps. The dataset is split into training, validation and test sets, and covers the 11 paintings viewed by participants during the visit at the museum. To ensure that the training set is representative, it includes at least one painting from each major category, i.e. portrait, genre scene with crowds, genre scene with few characters, pointillism, and surrealism. The test and validation sets are composed of the most frequently occurring types, i.e. genre scenes (with crowds and few characters), pointillism, and portrait to counterbalance the lack of paintings diversity. During model training, the paintings and their corresponding saliency maps are resized to 480×640 pixels to fit the UNETRSal input size. Fine-tuning the UNETRSal model on museum-based data enables to generate saliency maps that synthesize the different gaze patterns recorded on each painting.

4 Transformation Based on Saliency Maps

In order to create the new artworks, each painting is first normalized so that its pixel values range between 0 and 1. The saliency map simulated from UNETRSal is rescaled and resized to match the exact dimensions of the painting. To identify the key points of visual attention, the saliency map is smoothed using a Gaussian filter with $\sigma = 5$, and the local maxima of key regions are detected. To affect each pixel in the painting to a local maximum, an attention center is determined among these local maxima. This pixel-center association is based on both the spatial distance between the pixel and the various attention centers, and the saliency value of each attention center. Once the pixel-center association have been computed, the pixels coordinates are shifted towards its associated center in proportion to the local saliency value. As such, in highly salient regions, pixels are more strongly pulled towards their local center, creating an impression of expansion around these areas of interest. The applied deformation factor depends on a global parameter controlling the overall zoom intensity. This mechanism allows for precise enhancement of salient regions while maintaining visual coherence across the painting. The final artwork is obtained by using bilinear interpolation, which estimates color values at the new pixel positions.

5 Conclusion

This work introduces a novel interactive pipeline that bridges human gaze and machine vision to create perceptually grounded transformations of fine art paintings. The accompanying demo will allow visitors to observe paintings using eye-tracking glasses, generate personalized saliency-driven transformations, and compare them with the UNETRSal predictions. This setup highlights the interpretability, creativity, and emotional resonance of gaze-guided art generation, opening new perspectives for cognitive interaction in digital art installations.

Acknowledgments

This work was performed using the computing resources of CRI-ANN, and funded by GIP Millénaire Caen 2025 as part of the ABC project (<https://artbienetre cerveau.fr/>).

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